ALBALQA APPLIED UNIVERSITY

FACULTY OF ENGINEERING TECHNOLOGYS

LABORATORY

EXPERIMENT

Exp one ⊕ two

NAME: محمد محمود جميل

REG.NUMBEr

INSTRUCTOR: م. محمد المشاقبة

LAB TIME: 8:30 - 11:30 ح/أ

DATE: 17/7/2025

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Experiment (1):

- Step 7- Micro/win and S7-200 - simulation

* objectives:

- Convert a simple electrical ladder diaggram to a plc program
- know the difference between physical componints and program componants.
- To use Micro-win program and S7-200 simulation.
- Editing ladder diagram and test it.
- Monitoring status by S7-200 - simulator.

* Theory:

- Step 7- Micro/win makes programming of S7-200 easier.
- In step 7- Micro/win, click on new option in the project menu for new projett.
- After composing program, click the compile button on the taskbar. By clicking this Button, Software translates the program code Block into machine langaage for executing by the CPU
- A program cannot be downloaded to the CPU until it is compiled. If there is an error in your program, your program will not be compiled and Software will warn you about the errors in your program. After correcting errors, try to Compile again.

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- Inputs will be defined as [from $I_{0.0}$ to $I_{0.7}$] and [from $I_{1.0}$ to $I_{1.5}$]

- outputs will be defined as [from $Q_{0.0}$ to $Q_{0.7}$] and [from $Q_{1.0}$ to $Q_{1.1}$]

- Flags will be defined as [$M_{0.0}$, $M_{0.1}$, $M_{0.2}$, ...]

I : input , O : byte , Q : output , 0,1,2 → bit.

* Equipments :
- Micro - win program
- S7-200 simulation.

Program instruction:
- A network of LAD is arow of connected elements that form a complete circuit between the left and Right power rail. The left power rail represents the energized conductor whereas the Right power rail represents the return path conductor of the circuit.
- power flows from the left rail, through the closed contacts to the coils or boxes connected to the right power rail.
- you can then use the power flow to activate the outputs according to your program.

* The input can be repeated in ladder diagram.
* The output cann't be repeated.
* The S7-200 PLC consist of 14 inputs and 10 outputs.

For example:

$I_{0.1}$ → 0 → represent the no of Register that consist the Bit of input.

1 → represent the no of Bit that is inside the register.

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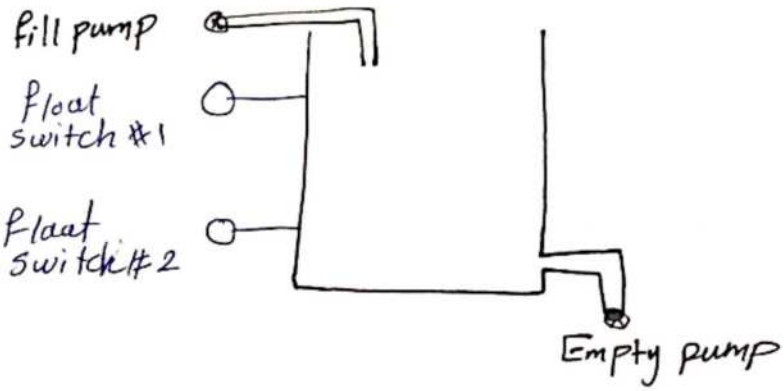
Experiment (2):

- Step 7 - Micro/win and S7-200 simulation problems.

* Problem:

In a recycle oil tank problem shown in the figue below,
- When the (Fill) button is pressed the Fill pump should run until float switch is activated on the stop button is pressed.

- when the (Empty) button is pressed and Float switch #2 is activated the empty pump should run until float switch #2 is de-activated on the stop button is pressed.



* write the ladder diagram and input/output table.

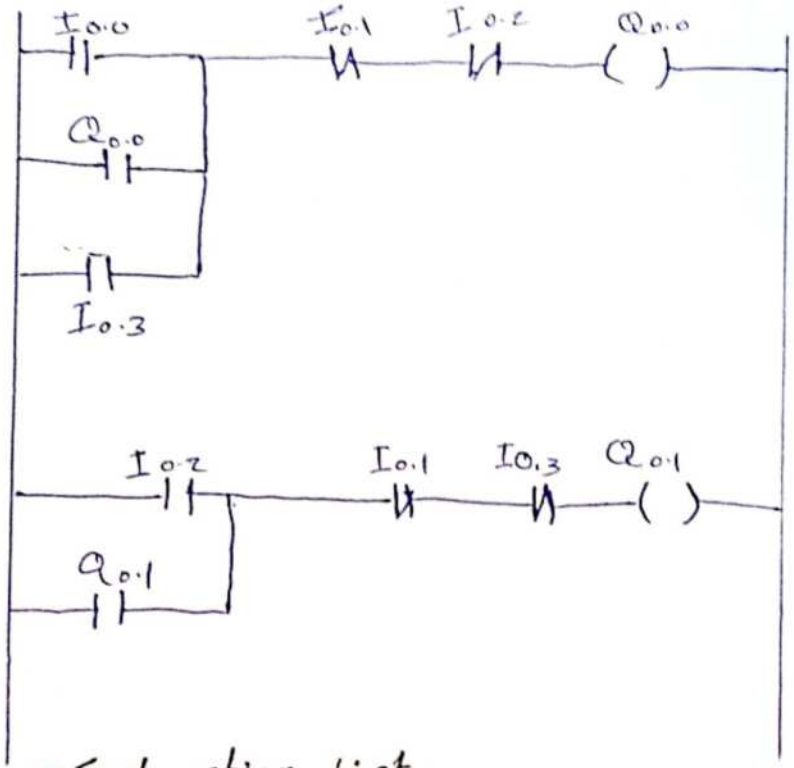
* write the program instruction list.

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Inputs	Out puts
Start → $I_{0.0}$	Fill pump → $Q_{0.0}$
Stop → $I_{0.1}$	Empty pump → $Q_{0.1}$ (valve)
Float switch # 1 → $I_{0.2}$	
Float Switch # 2 → $I_{0.3}$	

* Ladder diagram:



* Instraction list:

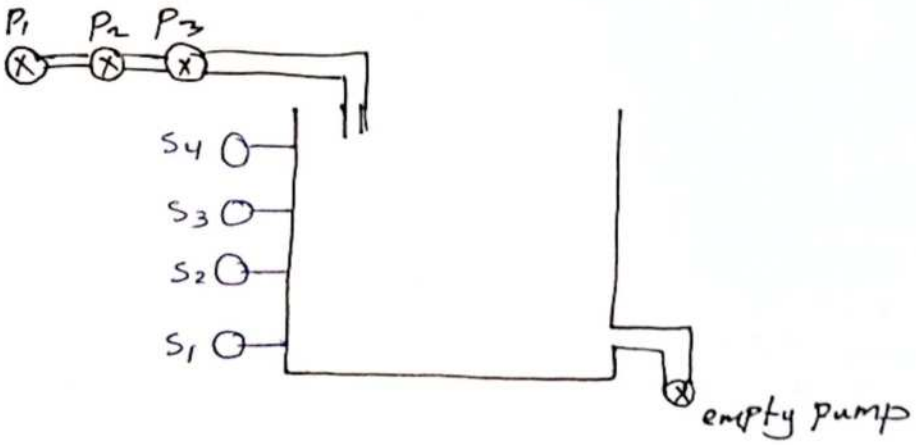
```
LD  I0.0
O   Q0.0
O   I0.3
A~  I0.1
A~  I0.2
=   Q0.0
LD  I0.2
O   Q0.1
AN  I0.1
AN  I0.3
=   Q0.1
```

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* problem 2:

- In figure below, when we pressed start button the Pill pump P_1, P_2, P_3 are run

- when S_2 is activated or the stop button pressed P_1 Stop.
- when S_3 is activated or the stop button pressed P_2 Stop.
- when S_4 is activated or the Stop button pressed P_3 stop.
- when S_4 is activated the empty pump should run until S_1 de-activated or stop button pressed.



* write ladder diagram and input/output table.

* write the program in intruction list.

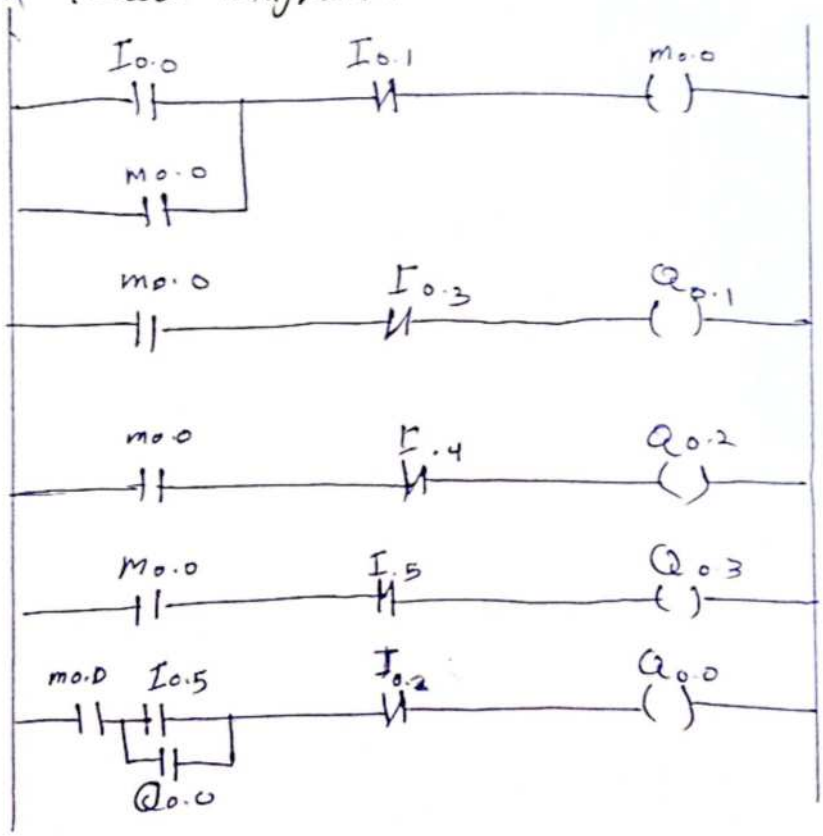
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Inputs	Output
Start → $I_{0.0}$	$P_1 \rightarrow Q_{0.1}$
Stop → $I_{0.1}$	$P_2 \rightarrow Q_{0.2}$
$S_1 \rightarrow I_{0.2}$	$P_3 \rightarrow Q_{0.3}$
$S_2 \rightarrow I_{0.3}$	empty pump → $Q_{0.0}$
$S_3 \rightarrow I_{0.4}$	
$S_4 \rightarrow I_{0.5}$	
masterCoil → $m_{0.0}$	

* Instraction List :

* ladder diagram :




```
LD   I0.0
O    m0.0
AN   I0.1
=    m0.0
LD   m0.0
AN   I0.3
=    Q0.1
LD   m0.0
AN   I0.4
=    Q0.2
LD   m0.0
AN   I0.5
=    Q0.3
LD   m0.0
LD   I0.5
O    Q0.0
ALD
AN   I0.2
=    Q0.0
```

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FACULTY OF ENGINEERING TECHNOLOGYS

LABORATORY

EXPERIMENT

Exp 3 ⊕ Exp 4

NAME: محمد محمود جبيل

REG.NUMBEP: [redacted]

INSTRUCTOR: م. محمد الشاعبة

LAB TIME: 8:30 - 11:30 ب/ا

DATE: 24/7/2025

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* Objective:

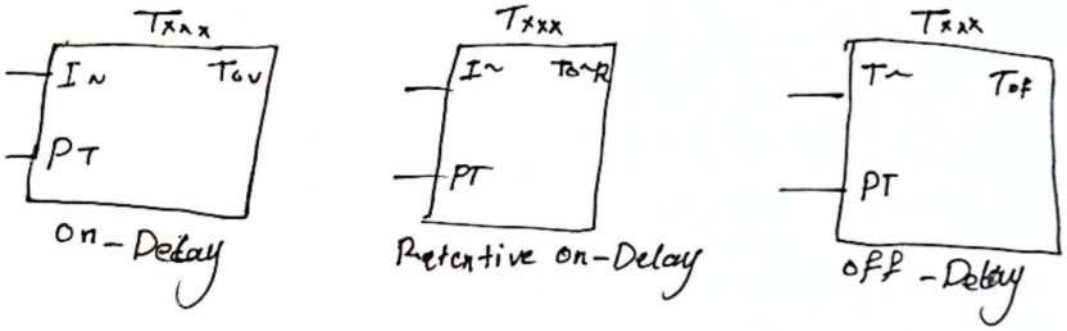
- To study the operation of different types of timers.
- To use the S7-200 timers in process control.

* Timer instruction:

- On-Delay Timer / Retentive on-Delay Timer.

- The on-Delay Timer (TON) and Retentive on-Delay Timer (TONR) instruction count time when the enabling input is on

- The time numbe (Txx) determines the resolution of the timer and the resolution is now shown in the instruction box.

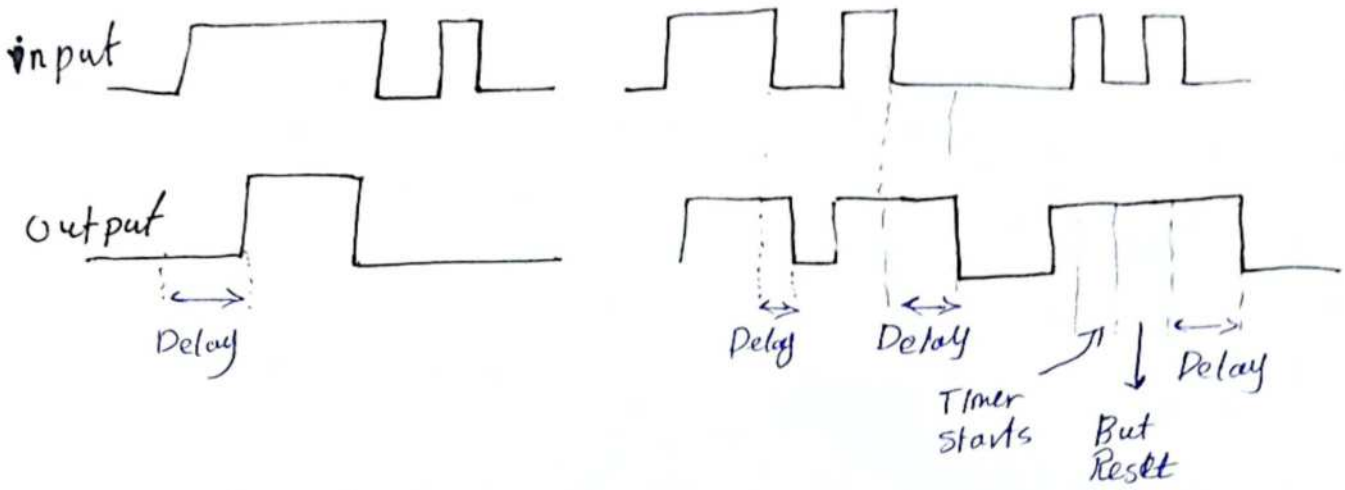


- off-Delay Timer:

- The off-Delay timer (TOF) is used to delay turning an output off for a fixed period of time after the input turns off.

* you can use a TON for timing a single interval.
* you can use a TONR for accumalating a number of timed intervals.
* you can use a TOF for exterding time past an off (or false) condition, such as for cooling a motor after it is turned off.

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* operation of the timer instruction :

Type	Currents = preset	State of enabling input	power cycle/first scan
TON	-Timer bit on -Current continues counting to 32,767	-ON : current value counts time - OFF : Timer bit off current value = 0	Timer bit off Current value = 0
TONR	- Timer bit on - current continues counting to 32,767	- ON : current value counts time - OFF : Time bit and current value maintain last state	Timer bit off -current value can be maintained.
TOF	-Timer bit OFF -current = preset, stops counting	ON: Timer bit on - current value = 0 - off : Time counts, after on-to-off transition	Timer bit off current value = 0

* The TON and TONR instructions count time when the enabling input is on. when the current value is equal to or grater than the preset time, the timer bit is on.
* The current value of TON timer is cleared when the enabling input is off, whereas the current value of the TONR timer is maintained when the input is off.
* you can use the TONR timer to accumulate time when the input turns on and off. Use the reset intruction (R) to clear the current value of the TONR
* Both the TONR timers continue counting after the preset is reached, and they stop counting at the maximum value of 32767.

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* The Tof instruction is used to delay turning an output off for a fixed period of time after the input turns off.
* when the enabling input turns on, the timer bit turns on immediately, and the current value is set to 0.
* when the input turns off, the timer counts until the elapsed time reaches the preset time.

* When the preset is reached, the timer bit turns off and the current value stops incrementing. However, if the input turns on again before the ToF reaches the preset value, the timer bit remains on.

* The enabling input must make an on-to-off transition for the ToF to begin counting time intervals.

* you can reset a TONR only by using the reset (R) instruction you can also use the reset instruction to reset any TON or TOF, the Reset instruction performs.

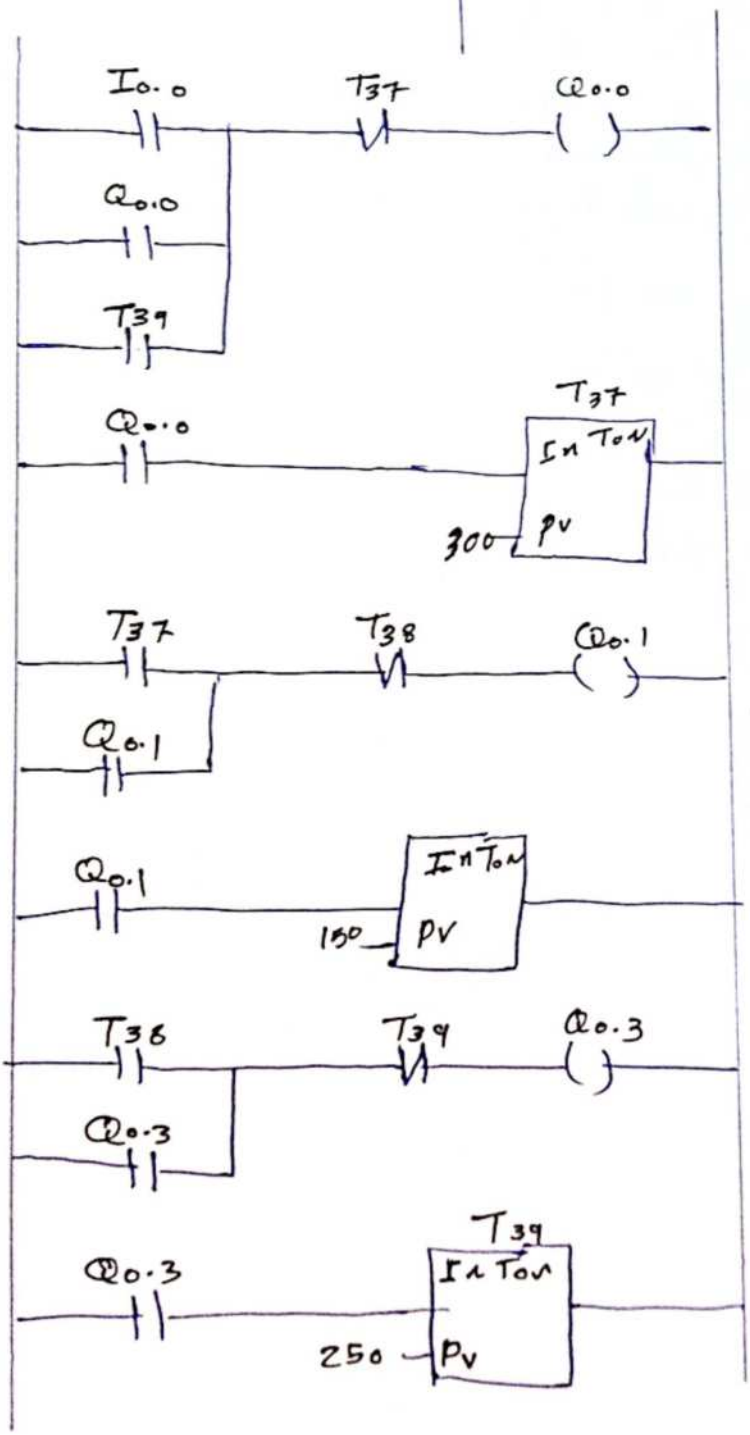
* Times numbers and Resolutions :

Timer type	Resolution	Maximum value	Timer number
TON TOFF	1ms	32 767	T_{32}, T_{96}
	10 ms	32 767	T_{33} - T_{36} , T_{97} - T_{100}
	100 ms	32 767	T_{37} - T_{63} , T_{101} - T_{255}
TONR	1ms	32767	T_0 , T_{64}
	10 ms	32 767	T_1 - T_4 , T_{65} - T_{68}
	100 ms	32 767	T_5 - T_{31} , T_{69} - T_{95}

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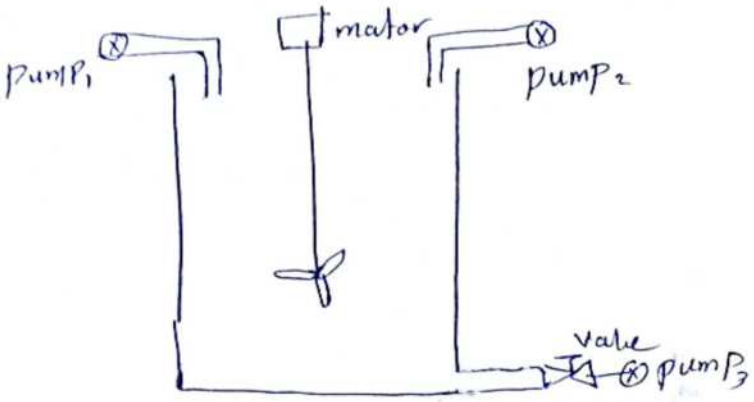
* The light traffic problem in one direction.

Input	Output
start → I0.0	Red light → Q0.0
	yellow light → Q0.1
stop → I0.1	green Light → Q0.2




```
LD  I0.0
O   Q0.0
O   T39
AN  T37
=   Q0.0
LD  Q0.0
TON T37,300
LD  T37
O   Q0.1
AN  T38
=   Q0.1
LD  Q0.1
TON T38,150
LD  T38
O   Q0.3
AN  T39
=   Q0.3
LD  Q0.3
TON T39,250
```

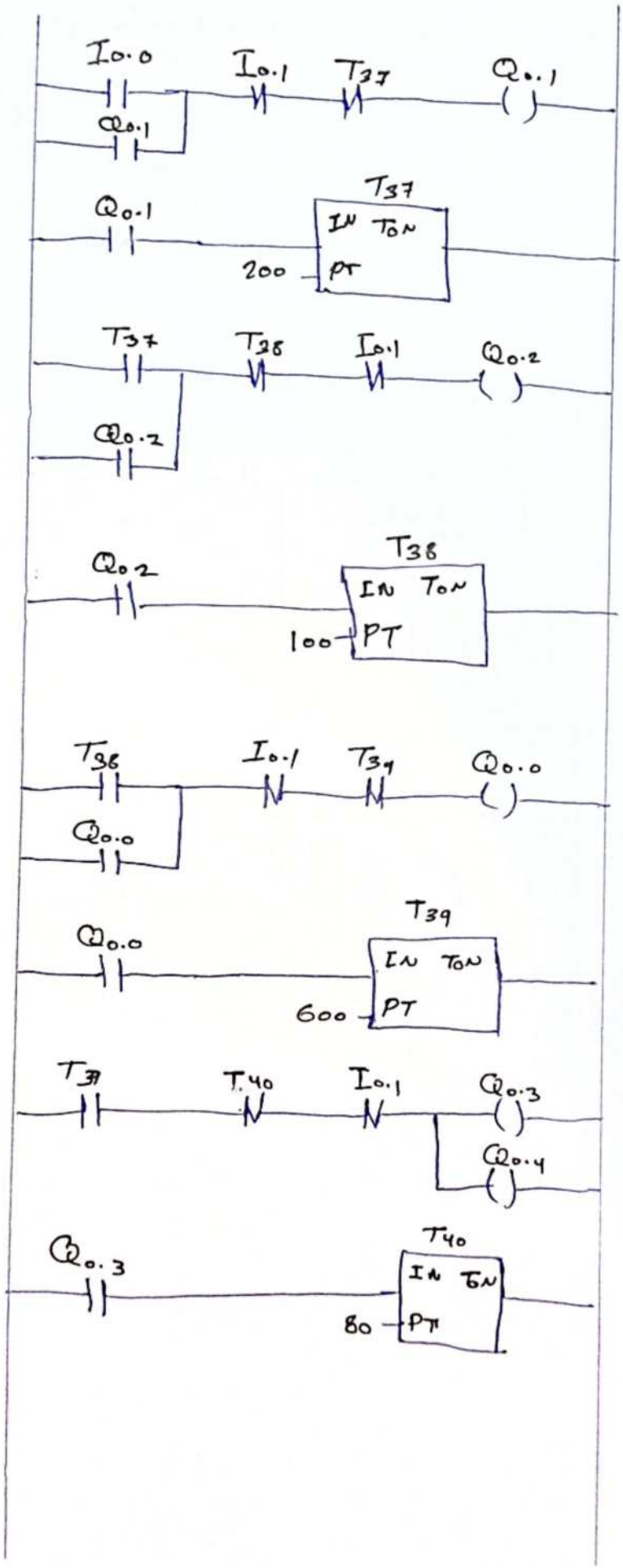
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Input	Output
start → I0.0	pump1 → Q0.1
stop → I0.1	pump2 → Q0.2
	pump3 → Q0.3
	motor → Q0.0
	Valve → Q0.4

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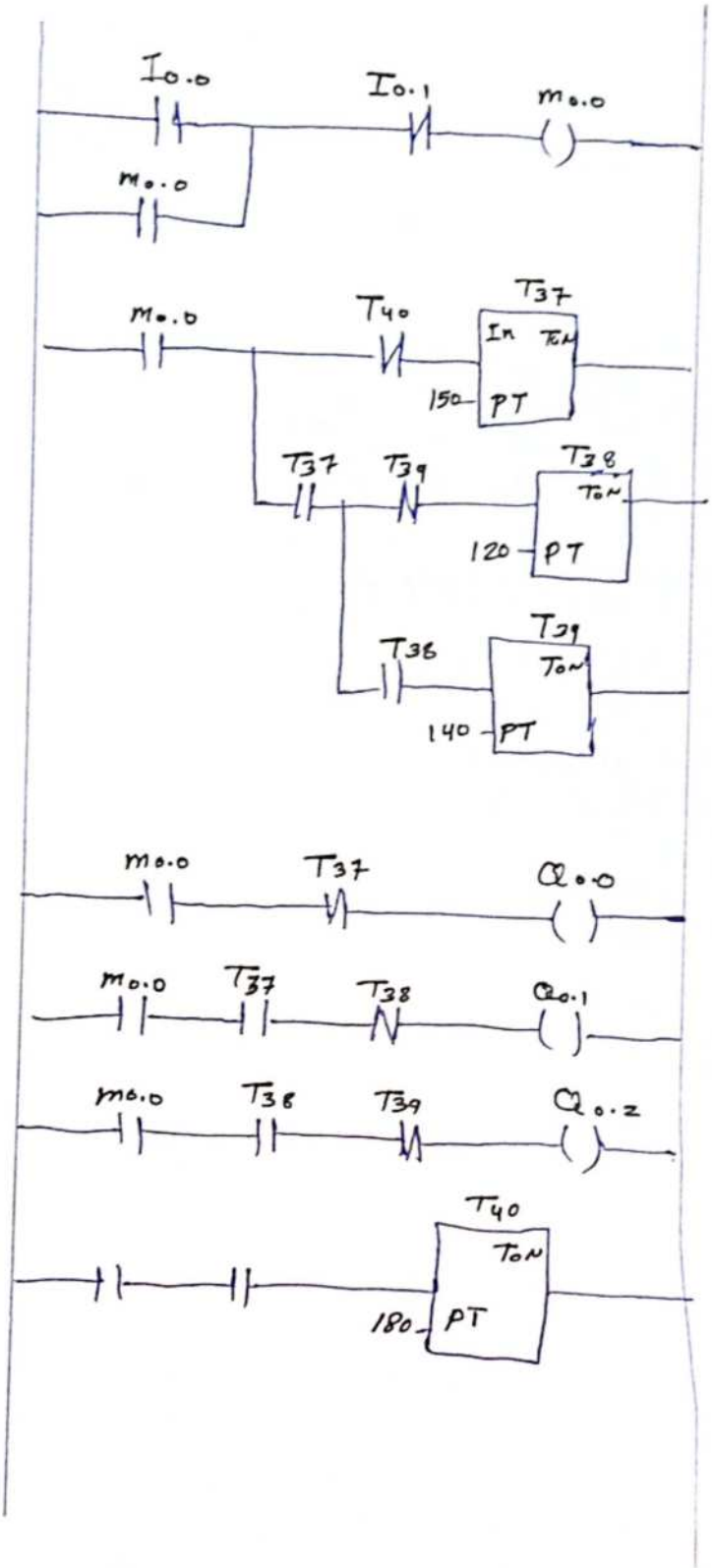
الممسوحة ضوئيا بـ CamScanner




```
LD   I0.0
O    Q0.0
AN   I0.1
AN   T37
=    Q0.1
LD   Q0.1
TON  T37, 200
LD   T37
O    Q0.2
AN   I0.1
AN   T38
=    Q0.2
LD   Q0.2
TON  T38, 100
LD   T38
O    Q0.0
AN   I0.1
AN   T39
=    Q0.0
LD   Q0.0
TON  T39, 600
LD   T39
AN   T40
AN   I0.1
=    Q0.3
=    Q0.4
LD   Q0.3
TON  T40, 80
```

الممسوحة ضوئيا بـ CamScanner

* Traffic Light with 2 direction.




```
LD   I0.0
O    m0.0
AN   I0.1
=    m0.0
LD   m0.0
AN   T40
TON  T37, 150
LD   m0.0
A    T37
AN   T39
TON  T38, 120
LD   m0.0
A    T37
A    T38
TON  T39, 140
LD   m0.0
AN   Q0.0
LD   m0.0
A    T37
AN   T38
=    Q0.0
LD   m0.0
A    T38
AN   T39
=    Q0.2
LD   m0.0
A    T37
TON  T40, 80
```

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FACULTY OF ENGINEERING TECHNOLOGYS

LABORATORY

EXPERIMENT

Exp 5

NAME: محمد محمود جميل

REG.NUMBER

INSTRUCTOR: م. محمد المشاقبة

LAB TIME: 8:30 - 11:30 أ/د

DATE: 30/7/2025

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* Experiment(5):

Communication PLC to PC:

1. Siamens PLC used in this progream: S7-200 CPU 212
   a- Connect part number PC/PPI cable from PC to siemens S7-200

2. Iunch siemens programming softwere.

3. Select the PLC-type.

4. Next, click on the Read PLC button to read your PLC type.

5. Click on the communications button and the setting should appear.

6. Assign the Remote address (in this testing, #2 is assigned)

7. Double Click

8. In the address path, select PC/PPI cable (PPI).
   a- click the properties button.

9. Local connection
   a- Assign the com pont, check your com pont befor assigning the number.

10. PPI:
    a-Assign the station address (#0 is assigned for this test)
    b_set the transmission rate at 9.6 Kbps.
    c- Highest node address is 31.
    d. click the ok button to exit from the propetics dialogue box.

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* Downloading the progrem to PLC:

- Click the Download icon on the toolbar or select the File > Download menu Command to download the program.

- Click ok on opened window to dowload the elements of the program to the S7-200.

- in the Download dialog box, follow all setting as shown in the image below and then click the ok button to download.

* Runing the PLC:

- For step 7- micro/WIN to place the S7-200 CPU is Run mode, the mode switch of the S7-200 must be set to TERM or RUN. when you place the S7-200 is RUN mode, the S7-200 executes the program.

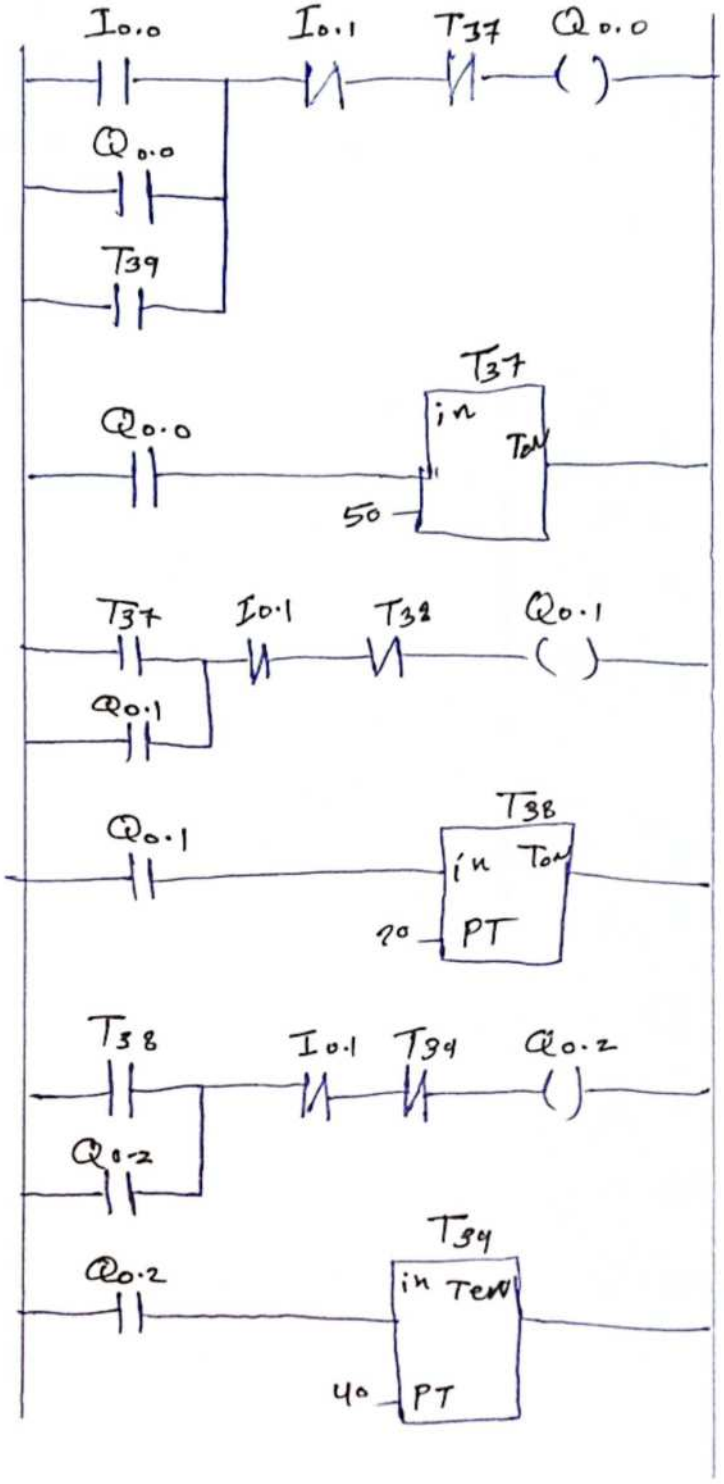
- Click the RUN icon on the toolbar or select the PLC > RUN mend command.

- Click ok to change the operating mode of the S7-200.

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* problem 1 :




```
LD  I0.0
O   Q0.0
O   T39
AN  I0.1
AN  T37
=   Q0.0
LD  Q0.0
TON T37,50
LD  T37
O   Q0.1
AN  I0.1
AN  T38
=   Q0.1
LD  Q0.1
TON T38,20
LD  T38
O   Q0.2
AN  I0.1
AN  T39
=   Q0.2
LD  Q0.2
TON T39,40
```

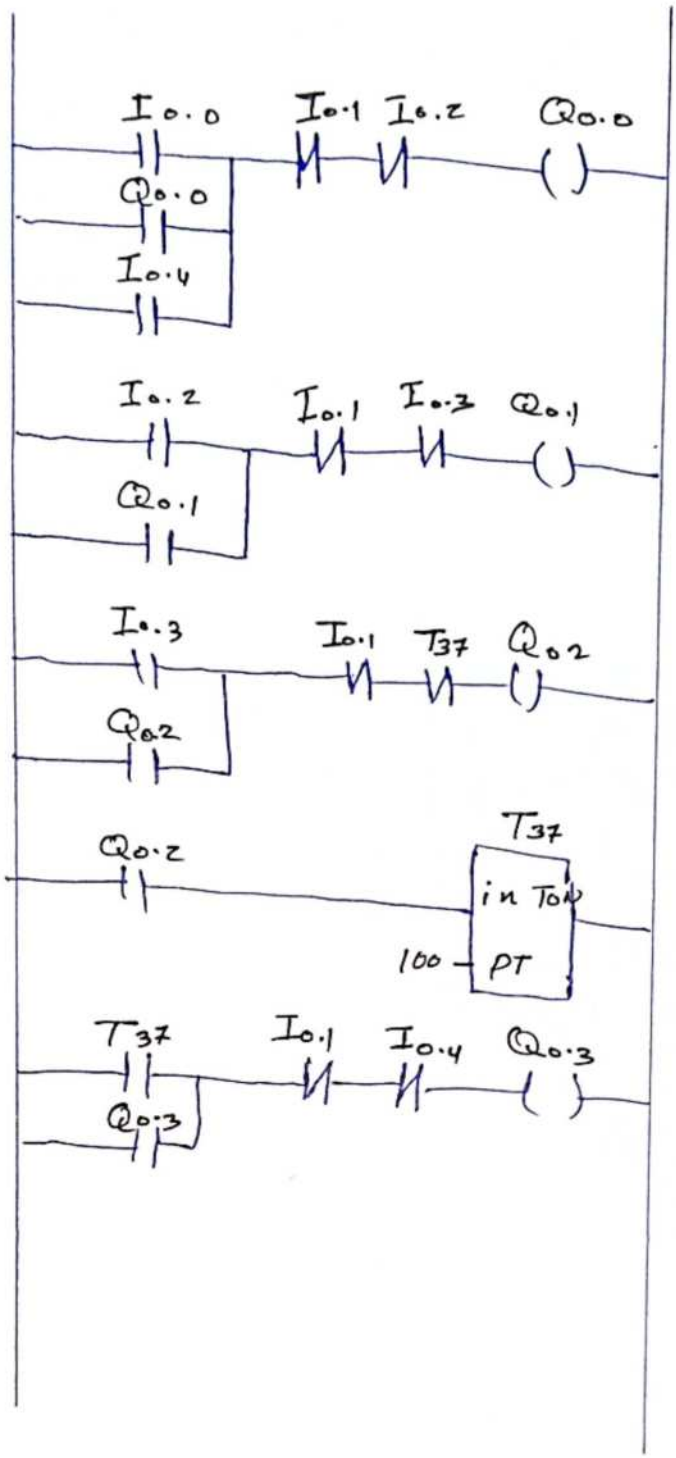
الممسوحة ضوئيا بـ CamScanner

Input	Output
Start → $I_{0.0}$	Red-light → $Q_{0.0}$
Stop → $I_{0.1}$	Yellow-Light → $Q_{0.1}$
	Green-Light → $Q_{0.1}$
	T_{ON} → T_{37}
	T_{ON} → T_{38}
	T_{ON} → T_{39}

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الممسوحة ضوئيا بـ CamScanner

* problem (2):




```
LD   I0.0
O    Q0.0
O    I0.4
AN   I0.1
AN   I0.2
=    Q0.0
LD   I0.2
O    Q0.1
AN   I0.1
AN   I0.3
=    Q0.1
LD   I0.3
O    Q0.3
AN   I0.1
AN   T37
=    Q0.2
LD   Q0.2
TON  T37, 100
LD   T37
O    Q0.3
AN   I0.1
AN   I0.4
=    Q0.3
```

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Input	Output.
Start → I0.0	Inlet valve 1 → Q0.0
Stop → I0.1	Inlet Valve 2 → Q0.1
level material A → I0.2	Agitator motor → Q0.2
level material B → I0.3	Outlet Valve → Q0.3
LLS → I0.4	

الممسوحة ضوئيا بـ CamScanner

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LABORATORY

EXPERIMENT

Exp 8

NAME: محمد محمود جبيل

REG.NUMBER: [redacted]

INSTRUCTOR: م. محمد المشاقبه

LAB TIME: 8:30 - 11:30 ص/أ

DATE: 17/8/2025

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الممسوحة ضوئيا بـ CamScanner

* PLC wiring Input and Output *

* Theory :

- Input and output :

• Input and output from a PLC are neccessary to monitor and control process.

• Both inputs and outputs can be categorized into two basic type : logical or continuous.

• Out puts from PLC s are often relays, but the can also be solide state electronics such as transistor for DC outputs or Triacs for AC outputs.

• Continuous outputs require special output cards with digital to analog converters.

• Input come from sensors that translate physical phenomena into electrical signals.

• PLC inputs must convent a variety of logic levels to the 5 vdc logic levels used on the Stalabus. Basically the circuits condition the input to drive an optocoupler this electrically isolates the extmal electrical Circuitry from the internal circuitry.

• PLC outputs must convent the 5 vdc logic levels on the PLC data bus to external voltage levels.

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* Inputs :

- In smaller PLCs the inputs are normally built in and are specified when purching the PLC.

- For langer PLCs the inputs ar purchased as moduls, or cards with 8 or 16 inputs of the same type on each card.

- Input devices such as (manual switches, push buttons, sensors)

- Typical ranges for input voltages : 12-24 Vdc / 5 Vdc (TTL)
  100-120 VAc / 200-280 Vac
  10-60 Vdc / 48 Vdc
  12-24 Vac/dc / 24 Vac

* Outputs :

- External power supplies are connected to the output card and the card will switch the power on or off for each output.

- Output devices such as (contactors and relay, motor, pump, ---)

- Typical output voltages are : 120 Vac / 24 Vdc / 12-48 Vac
  12-48 Vdc / 5 Vdc (TTL) / 230 Vac

- Relays are the most flexible output devices. They are capable of switching both Ac and Dc outputs.

- Transistor are limited to DC outputs.
- Triacs are limited to AC outputs.
- Transistor and triac outputs are called switched outputs.

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* Analog I/O:

- In the early days of programmable logic controllers processor speed and memory were too limited to support anything but discrete (on/off) control functions.

- Modern PLC technology, though, is powerful enough to support the measurment, processing, and output of analog (continuously variable) signals.

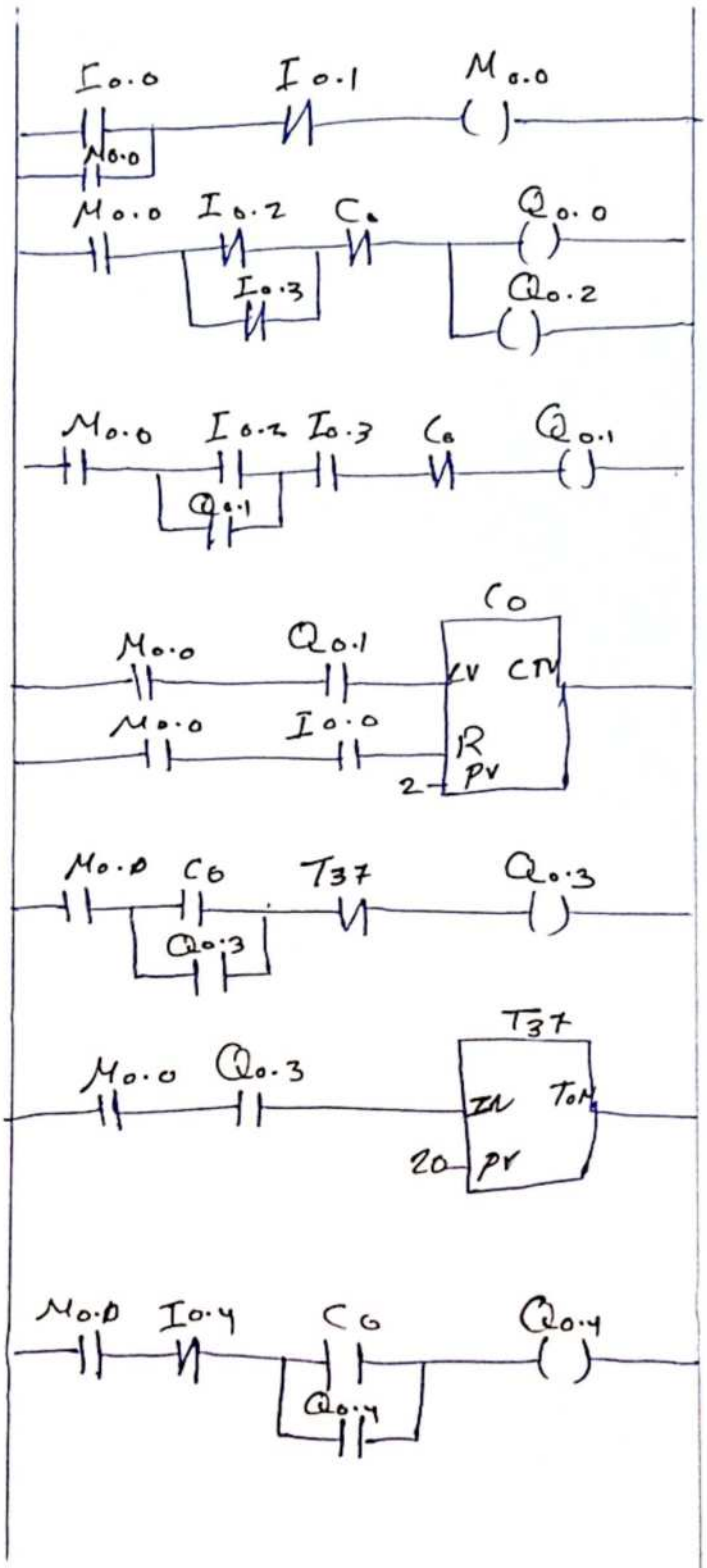
- insid every analog input module is an ADC.
- every analog output module contains a DAC.

- Typical analogue signals from (0-10 V_{DC}) or (4-20 mA) they are used to represent changing values such as voltage, current, speed, temperature, weight, level, Thermo couple, RTD and strain gauge.

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CamScanner الممسوحة ضوئيا بـ

problem:




```
LD   I0.0
O    M0.0
AN   I0.1
=    M0.0
LD   M0.0
LDN  I0.2
ON   I0.3
ALD
AN   C0
=    Q0.0
=    Q0.2

LD   M0.0
LD   I0.2
O    Q0.1
ALD
A    I0.3
AN   C0
=    Q0.1
LD   M0.0
=    I0.0
CTU  C0,2
LD   M0.0
LD   C0
O    Q0.3
AN   T37
=    Q0.3
LD   M0.0
A    Q0.3
TON  T37,20
LD   M0.0
AN   I0.4
LD   C0
O    Q0.4
ALD
=    Q0.4
```

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Input	output.
Start → I0.0	MV1 → Q0.0
Stop → I0.1	MV2 → Q0.1
TBL1 → I0.2	mixing motor → Q0.2
TBL2 → I0.3	TON → T37
Reset → I0.4	CTV → C0
	Buzzer → Q0.3
	LAMBA → Q0.4

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LABORATORY

EXPERIMENT

EXP 9 + EXP 10

NAME: محمد محمود جميل

REG.NUMBER: 3

INSTRUCTOR: م. محمد الحشاقبه

LAB TIME: 8:30 - 11:30 ص/أ

DATE: 21/8/2025

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* LoGo logic *

* Theory :

- The LoGo logic module has an integrated controller with operator control and display unit.

- You can use the operator control and display unit of LoGo to create and edit programs and execute system functions.

- You can read in external programs from a program module via the Ethernet interface or a PC cable with the LoGo soft comfort programing software. You can use LoGo soft comfort not only to create a program but also to simulate your circuit on the computer and to output over view diagrams to a printer.

- This PLC has 8 inputs, 4 outputs, a LCD display screen and a key pad with 6 keys (UP, Down, Right, left, Esc and Ok)

- The display screen and keypad presented with the PLC and the viewing elements for states, defaults and other massages.
- They made, data and time, program version or the number of cycles are displayed.

* Keys ▲ ▼ are used to change screen vertically.
* Keys ◀ ▶ are used to change screen horizontally.

* Esc key is to be used to go back to the previous menu level in case a mistake in entering accurred at one stage.

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* Constant and connectors (Co):
- Constants and connectors represent inputs, outputs, flags and constant voltage levels.

* Inputs:
• Digital inputs:
- Digital inputs are identified with an I ($I_1, I_2, \ldots$)

• Analog inputs:
- we can use I_7 and I_8 inputs to analog signal, which can also be programed for use as $AI_1 \ldots AI_8$ inputs, analog flags $AM_1 \ldots AM_6$, and analog output AQ_1 and AQ_2.

* Outputs:
• Digital outputs.
- Digital outputs ar identified by the character Q ($Q_1, Q_2, \ldots$)

• Analog outputs:
- Analog outputs are identified by the lettars AQ.
Tow analog outputs are available, namely AQ_1 and AQ_2.

* Flage blocks:
- Flage blocks are identified by the lettars M or AM

- These are virtual outputs, which output the value of their inputs. LOGO: provides 24 digital flags $M_1 \ldots M_{24}$ and 6 analog flags $AM_1 \ldots AM_6$

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** General Function (GF):

- Basic functions represent simple logical elements of Boolean algebra. The GF list contains the basic function blocks you can use for your circuit program.

** Special Functions List-(SF):

- when you create your circuit program in LoGo! You find the special function block in SF List.

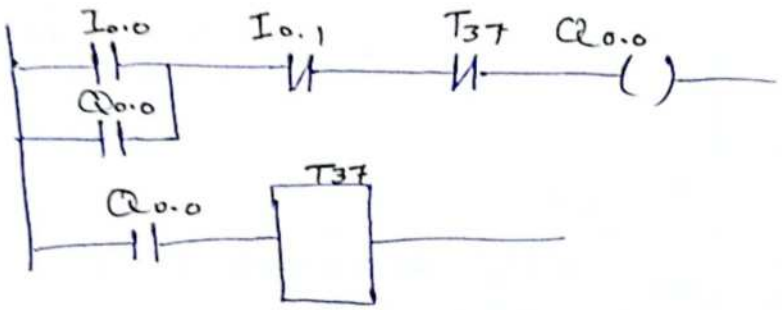
the following some of SFs:

- Analog comparator.
- Analog amplifier.
- Yearly timer.

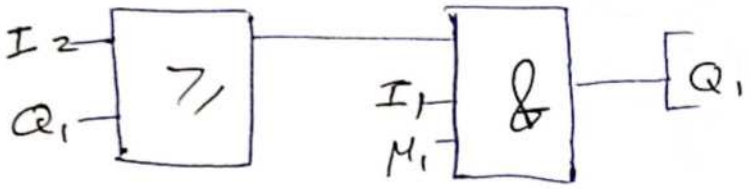
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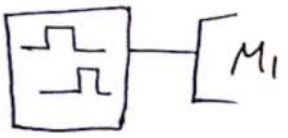
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* problem :



* Equevelant functional Block :



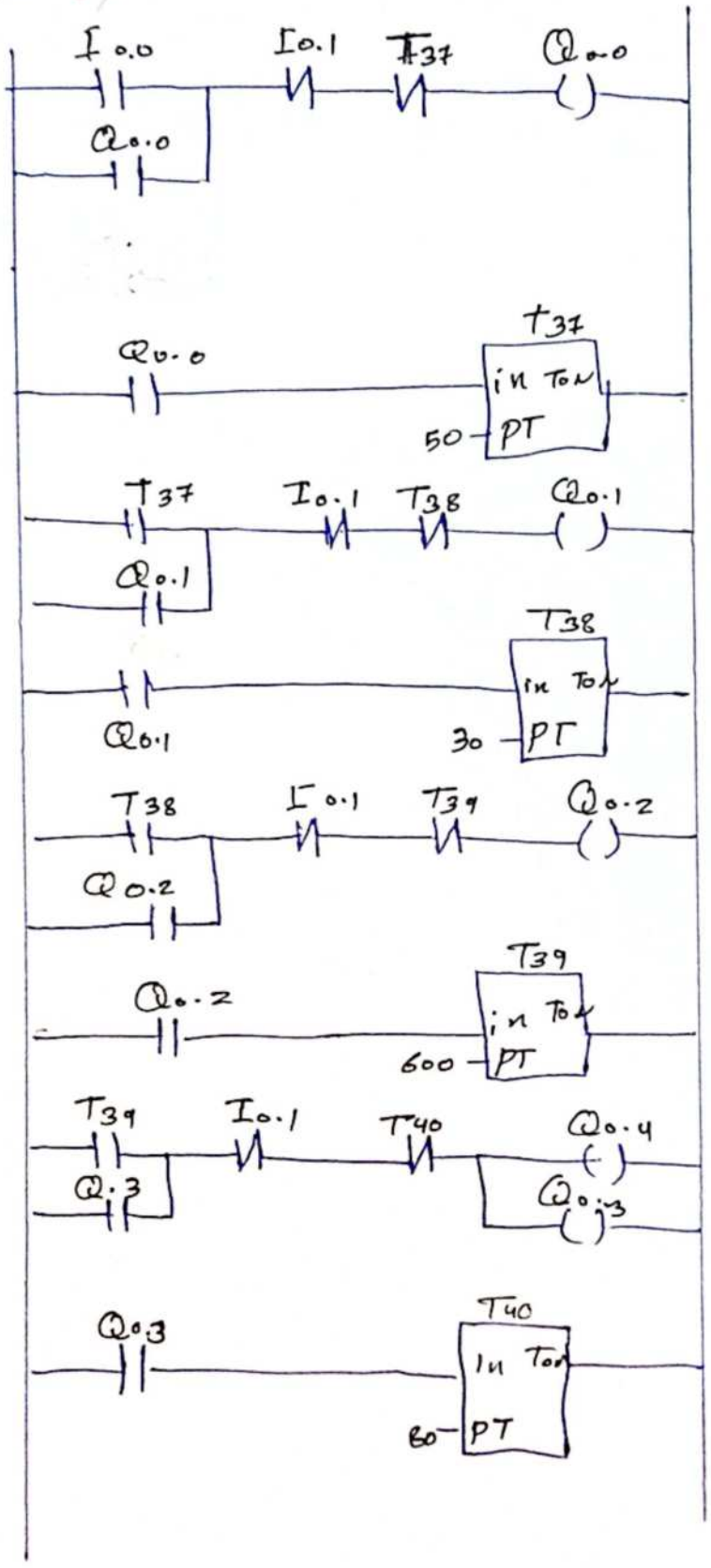


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Input	output
start → I0.0	Red → Q0.0
stop → I0.1	yellow → Q0.1
	Green → Q0.2
	TON → T37
	TON → T38
	TON → T39

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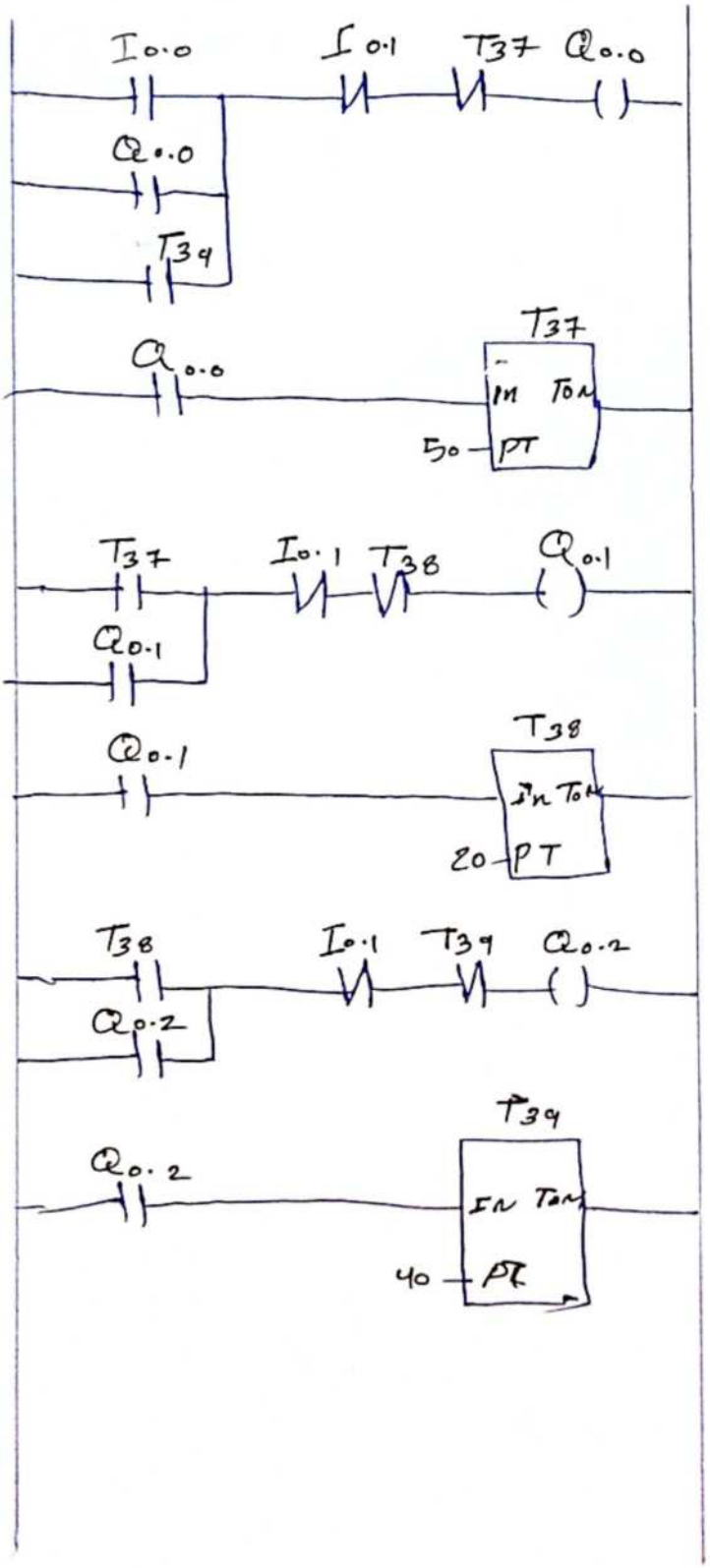
* problem (1):




```
LD   I0.0
O    Q0.0
AN   I0.0
AN   T37
=    Q0.0
LD   Q0.0
TON  T37,50
LD   T37
O    Q0.1
AN   I0.1
AN   T38
=    Q0.1
LD   Q0.1
TON  T38,30
LD   T38
O    Q0.2
AN   I0.2
AN   T39
=    Q0.2
LD   Q0.2
TON  T39,600
LD   T39
O    Q0.3
AN   I0.3
AN   T40
=    Q0.4
=    Q0.3
LD   Q0.3
TON  T40,80
```

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* problem (2) :



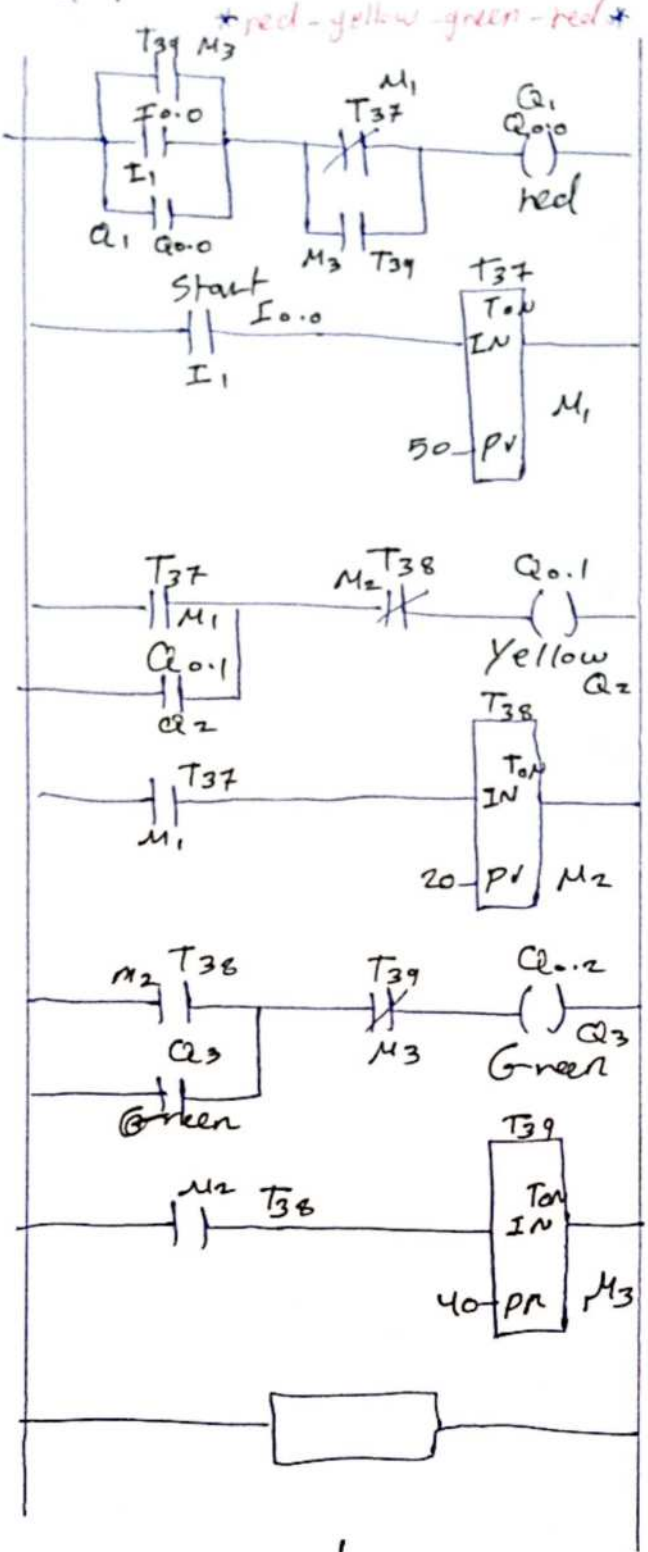

```
LD    I0.0
O     Q0.0
O     T39
AN    I0.1
AN    T37
=     Q0.0
LD    Q0.0
TON   T37,50
LD    T37
O     Q0.1
AN    I0.1
AN    T38
=     Q0.1
LD    Q0.1
TON   T38,20
LD    T38
O     Q0.2
AN    I0.1
AN    0.2
=     Q0.2
LD    Q0.2
TON   T39,40
```

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* Problem (2) :

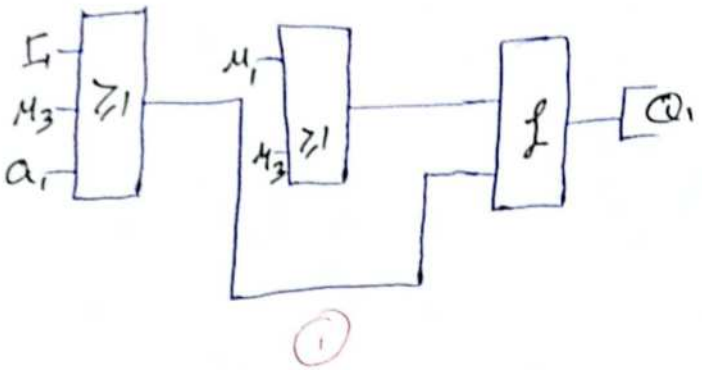
* red - yellow - green - red *



Inputs	Outputs
start I0.0	red → Q0.0
	Yellow → Q0.1
	Green → Q0.2

3 timer

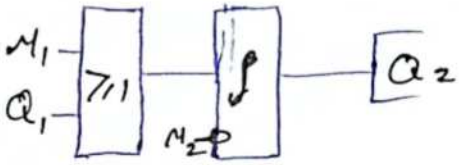
* Functional block.



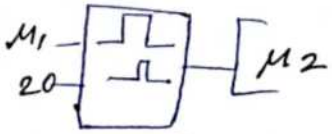
(1)



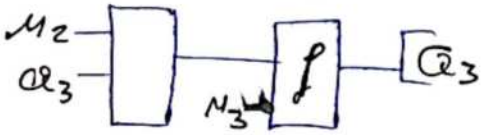
(2)



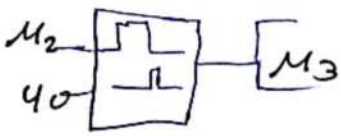
(3)



(4)



(5)



(6)

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